

TRANSPORTATION & LOGISTICS

ANALYTICS DASHBOARD

Exploring shipment delays, operational bottlenecks, and performance drivers through data.

6,773 SHIPMENTS | MARCH 2019 - SEPTEMBER 2020

PYTHON | SQL | POWER BI



WHY ARE SHIPMENTS DELAYED?

Operational data was scattered across suppliers, vehicles, routes, GPS providers, and delivery events. Without one connected view, teams could see delays but not where the operation was breaking down.



WHERE?

Where do delays concentrate?

WHO?

Who drives the greatest risk?

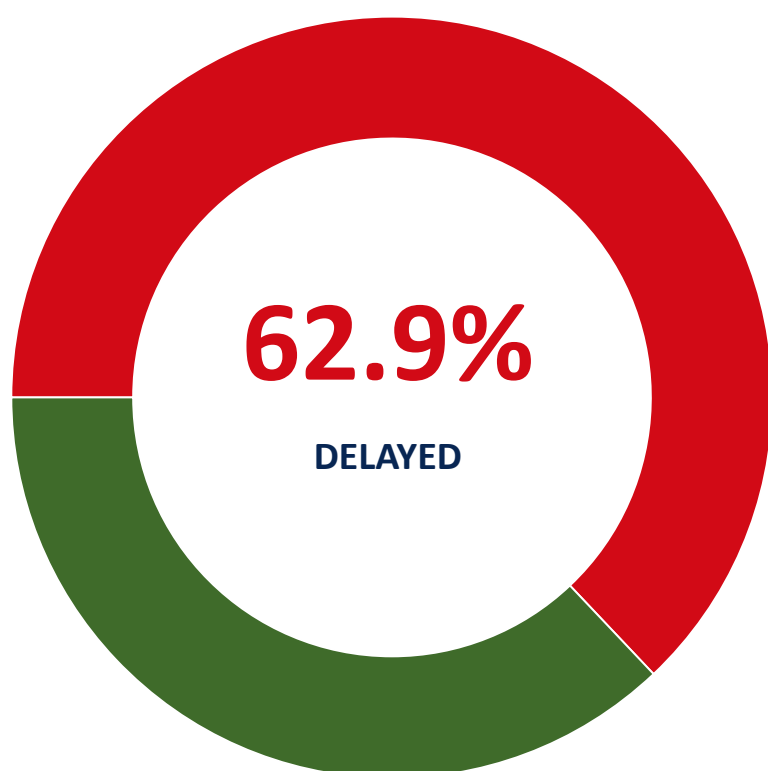
WHAT NEXT?

What should change operationally?

62.9% OF SHIPMENTS WERE DELAYED

The data revealed a systemic delivery performance problem, not isolated exceptions.

DELIVERY STATUS



● Delayed 4,263 | 62.9%

● On-Time 2,510 | 37.1%

DELAY DEFINITION

Actual ETA > Planned ETA

WHAT THIS MEANS

Nearly 2 in every 3 shipments missed the planned ETA.

SYSTEM-WIDE ISSUE

4,263 of 6,773 cleaned shipment records were classified as delayed.

NOT ISOLATED EXCEPTIONS

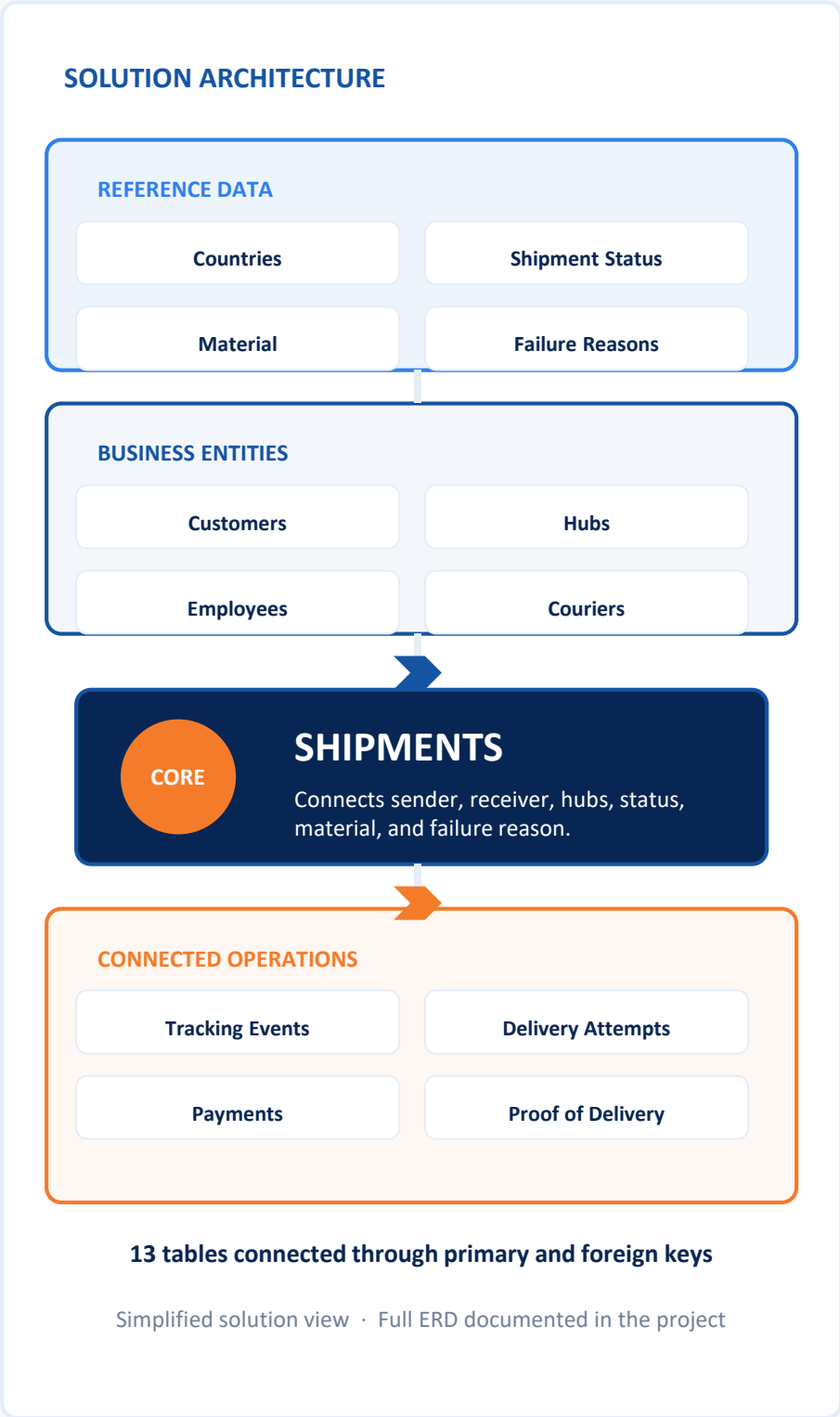
Delay patterns shifted across time, distance, suppliers, vehicle types, and GPS providers.

NEXT

Structure the operation for deeper SQL analysis.

A CONNECTED DATABASE FOR END-TO-END SHIPMENT CONTROL

The solution transforms fragmented logistics records into a structured SQL Server data foundation.



I designed a relational SQL Server database to turn disconnected records into one traceable logistics system.

13
RELATIONAL TABLES

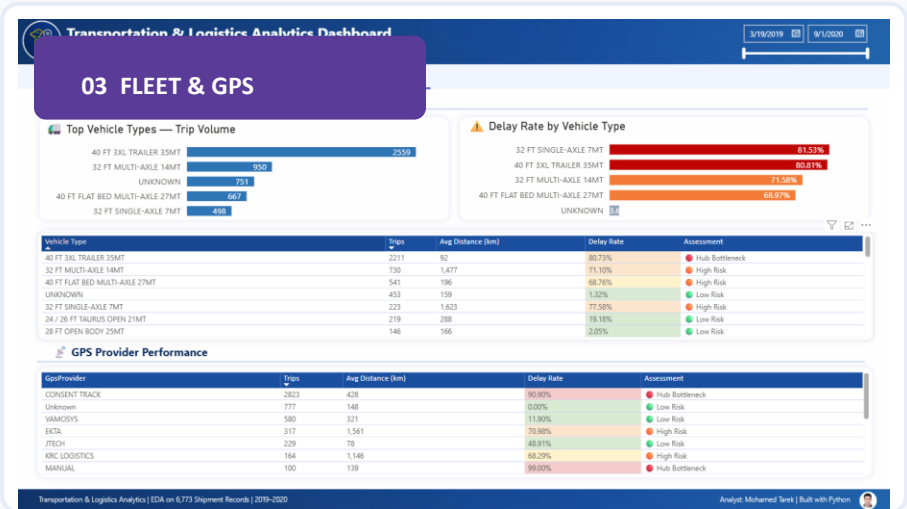
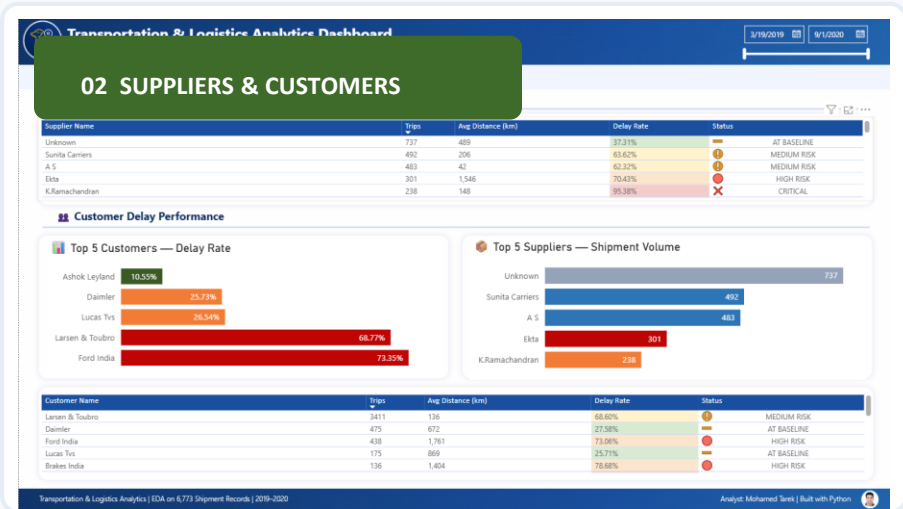
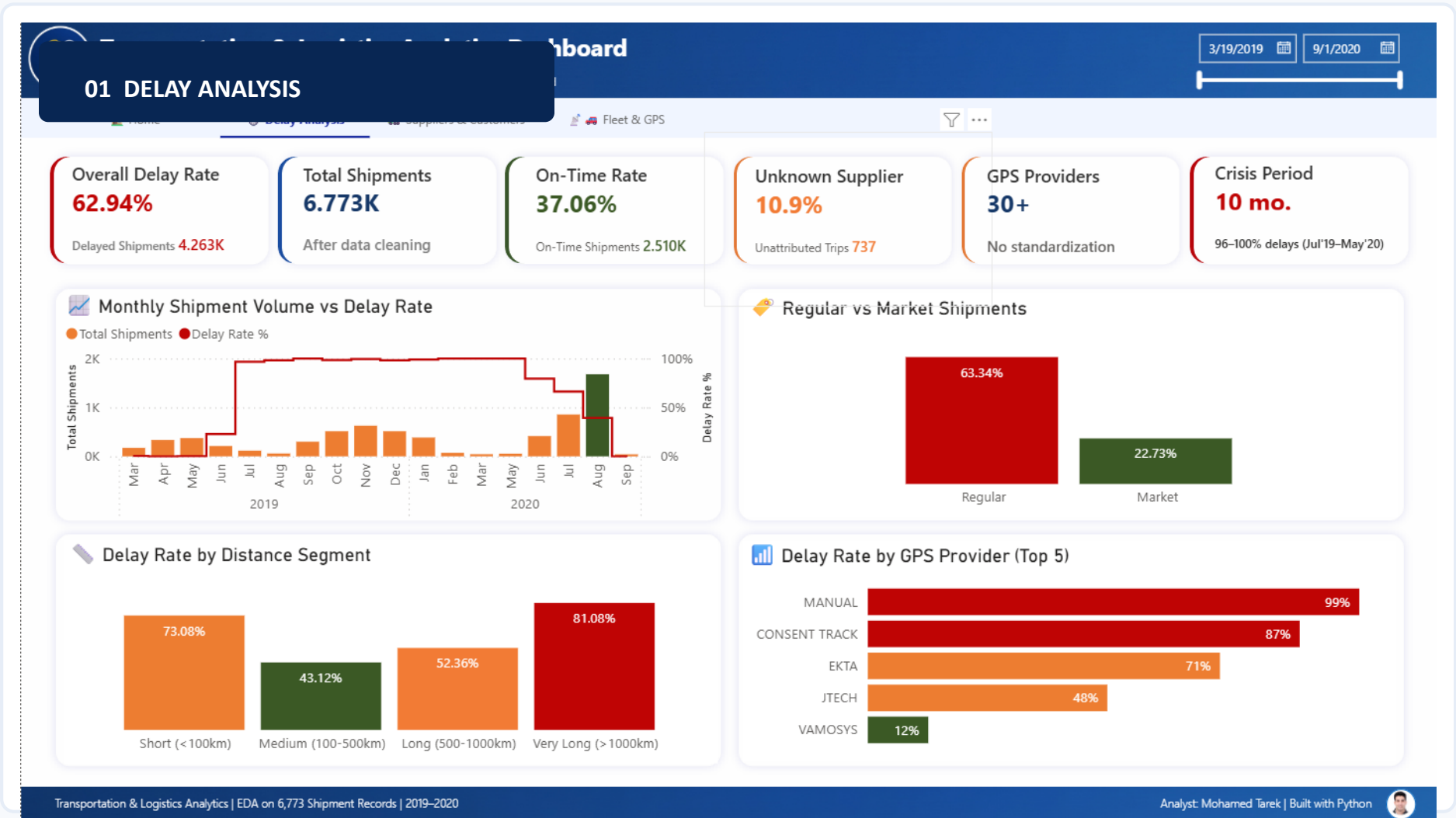
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ANALYTICAL QUERIES

- 01 SINGLE SOURCE OF TRUTH**
Customers, hubs, reference data, and shipments are connected through validated relationships.
- 02 END-TO-END TRACEABILITY**
Every tracking event, delivery attempt, payment, and proof of delivery remains linked to its shipment.
- 03 DECISION-READY FOUNDATION**
Ten reusable queries expose delays, courier performance, hub throughput, stuck shipments, POD, and revenue.

RESULT
A trusted SQL foundation for operational monitoring and Power BI.

ONE DASHBOARD. THREE OPERATIONAL VIEWS.

Interactive pages move from the overall delay problem to the supplier, customer, vehicle, and GPS drivers behind it.



FROM SQL MODEL TO INTERACTIVE MONITORING

6,773 SHIPMENTS · 32 FEATURES · MAR 2019–SEP 2020

THE DELAY PROBLEM HAS CLEAR OPERATIONAL DRIVERS

The risk is concentrated across time, distance, tracking quality, vehicle type, and partner performance.

CRISIS MONTHS

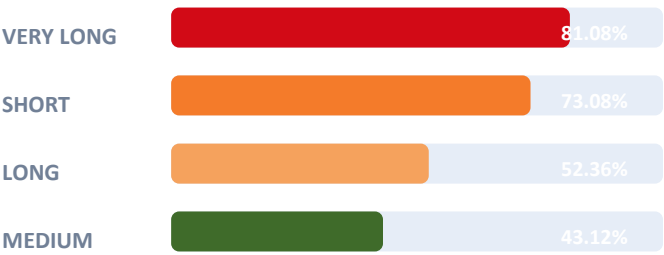
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monthly delay rate reached 96–100%

JUL '19 – MAY '20
WINDOW

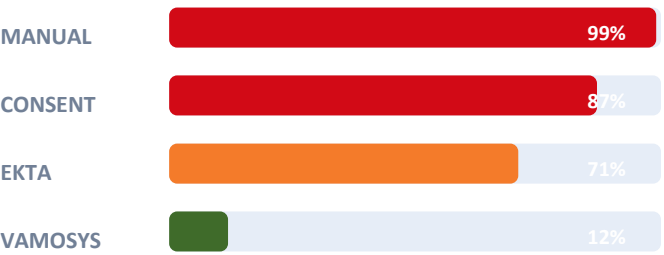
01 DISTANCE RISK

Very long and short routes show the highest delay rates.



02 GPS PROVIDER GAP

Tracking-provider performance varies dramatically.



03 VEHICLE PRESSURE

81.53%

32 FT SINGLE-AXLE 7MT

80.81%

40 FT 3XL TRAILER 35MT

2,559 trips on the highest-volume 40 FT 3XL trailer

04 PARTNER HOTSPOTS

SUPPLIER

95.38%

K. Ramachandran

CUSTOMER

73.35%

Ford India

Delay risk is concentrated, not evenly distributed across partners.

TAKEAWAY

A single blanket fix will not work—the response must target routes, tracking providers, vehicle types, and high-risk partners.

TURN INSIGHTS INTO AN OPERATING PLAN

Each action targets a bottleneck exposed by the EDA, SQL analysis, and Power BI dashboard.

PRIORITY ACTIONS

01

CONTROL HIGH-RISK ROUTES
81.08% very long · 73.08% short

Create route-specific service rules, capacity plans, and exception alerts instead of one operating standard for every distance band.

02

STANDARDIZE GPS DATA
MANUAL 99% · CONSENT TRACK 87%

Reduce manual tracking, standardize provider names, and monitor GPS completeness and provider-level delay performance.

03

INTERVENE BY VEHICLE TYPE
81.53% single-axle · 80.81% 3XL trailer

Audit dispatch, maintenance, load planning, and route assignment for the vehicle types carrying the highest delay exposure.

04

ENFORCE PARTNER ACCOUNTABILITY
95.38% supplier · 73.35% customer

Run root-cause reviews with high-risk partners, define service-level ownership, and resolve the 737 unknown-supplier records.



WHAT SUCCESS LOOKS LIKE

- ✓ Lower delay rate
- ✓ Higher on-time rate
- ✓ Cleaner GPS and supplier data
- ✓ Fewer failed or stuck shipments

OPERATING PRINCIPLE

Use the dashboard as a recurring management system, not a one-time report.

EXPLORE THE FULL END-TO-END PROJECT

From raw logistics data to business recommendations, the complete analytical workflow is structured for GitHub.



GITHUB

MOHAMEDTAREK00212 /

Transportation-Logistics

Transportation & Logistics Analytics
End-to-End Data Analytics and BI Case Study

VIEW THE PROJECT

github.com/Mohamedtarek00212/Transportation-Logistics

LINKEDIN | mohamedtarek-aboelkhir

LIVE POWER BI | INTERACTIVE DASHBOARD

END-TO-END PROJECT SCOPE

01

CLEANED DATASET
Validated analysis-ready records

02

EDA NOTEBOOK
Cleaning, exploration, and findings

03

SQL DATABASE
ERD, 13 tables, and 10 queries

04

QUERY OUTPUTS
Evidence from analytical SQL

05

POWER BI
Three interactive dashboard views

06

REPORT & STORY
Arabic report and business narrative

PROJECT JOURNEY

DATA

>

EDA

>

SQL

>

POWER BI

>

DECISIONS